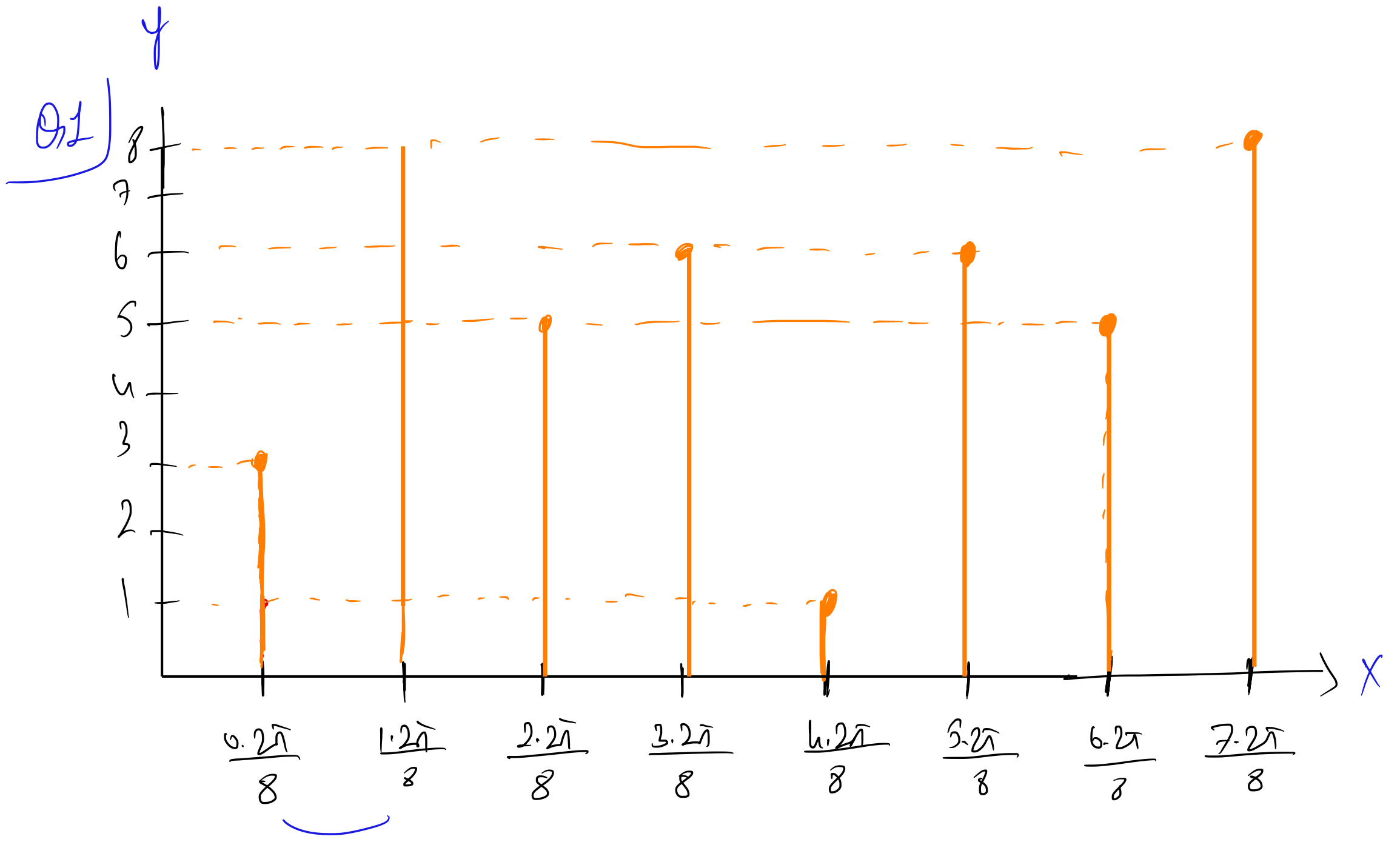
Q1)



X ekseni each step $\Rightarrow w = \frac{2\pi k}{8}$ $(k=0, \; k=7)$

y ekseni $|\hat{X}[k]|$ değerleri

Note) The figure shows the magnitude spectrum of periodic sequence $\tilde{x}[n]$, with period 8. The DTFT $\tilde{X}(e^{jw})$ has been sampled at $w = \frac{2\pi k}{8}$ and plotted as a discreate magnitude spectrum.

Q2)

a)

Lenght $(y[n])$ = Lenght $(x[n])$ + Lenght $(h[n])$ − 1

$x[n] \Rightarrow (n = 1 \ldots 39)$ tanımlı

$h[n] \Rightarrow$ 8 noktada tanımlı $(0-7)$ ama en etkilisi 2-5 arası k nok.

Support:

$x[n]$ destek : $[1, 39]$

$h[n]$ destek : $[2, 5]$

Support $(y[n]) = [1+2, 39+5] = [3, 44]$

Answer: $y[n]$ is nonzero for $3 \leq n \leq 44$

b)

DFT lenght $\geq$ Lenght$(x[n])$ + Lenght $(h[n])$ − 1

But
- lenght $(x[n]) = 39$
- Lenght $(h[n]) = 8$

gerekli min. uzunluk $= 39 + 8 = 46$

But we just using 40 point DFT. So not the same each other $y[n]$ and $y_2[n]$.

the region where its equal to:

$n = 0$ to (DFT_lenght − h_lenght) $= 40 - 8 = 32$

$y_2[n] = y[n]$ for $0 \leq n \leq 32$

$\boxed{y_2[n] = y[n] \text{ for } 3 \leq n \leq 32}$

c)

1)

$x[n]$ = lenght of each block 20

$h[n]$ = only first 6 values (because effective $2 \leq n \leq 5$)

DFT lenght $\geq$ Lenght of x block + Legth of h − 1

Min DFT length $= 20 + 6 - 1 = 25$

2)

- $x[n]$ $1 \leq n \leq 39 \rightarrow$ toplam 39
- her blok 20 uzunlukta

$\frac{39}{20} = 1{,}95 \Rightarrow$ 2 blok gerekir.

1. blok $x[1] - x[20]$
2. blok $x[21] - x[39]$

2 block needed.

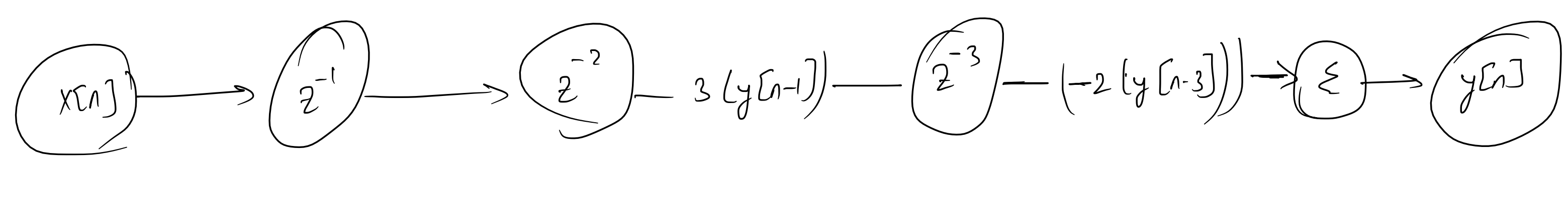
3)

Answer) The output of first segment convolution is written at $n = 0$ to 24.

The output of second segment is added starting from $n = 20$, overlapping with tail of first. The two partial results are summed at overlapping indices to reconstructs $y[n]$.

Q3) a

1)



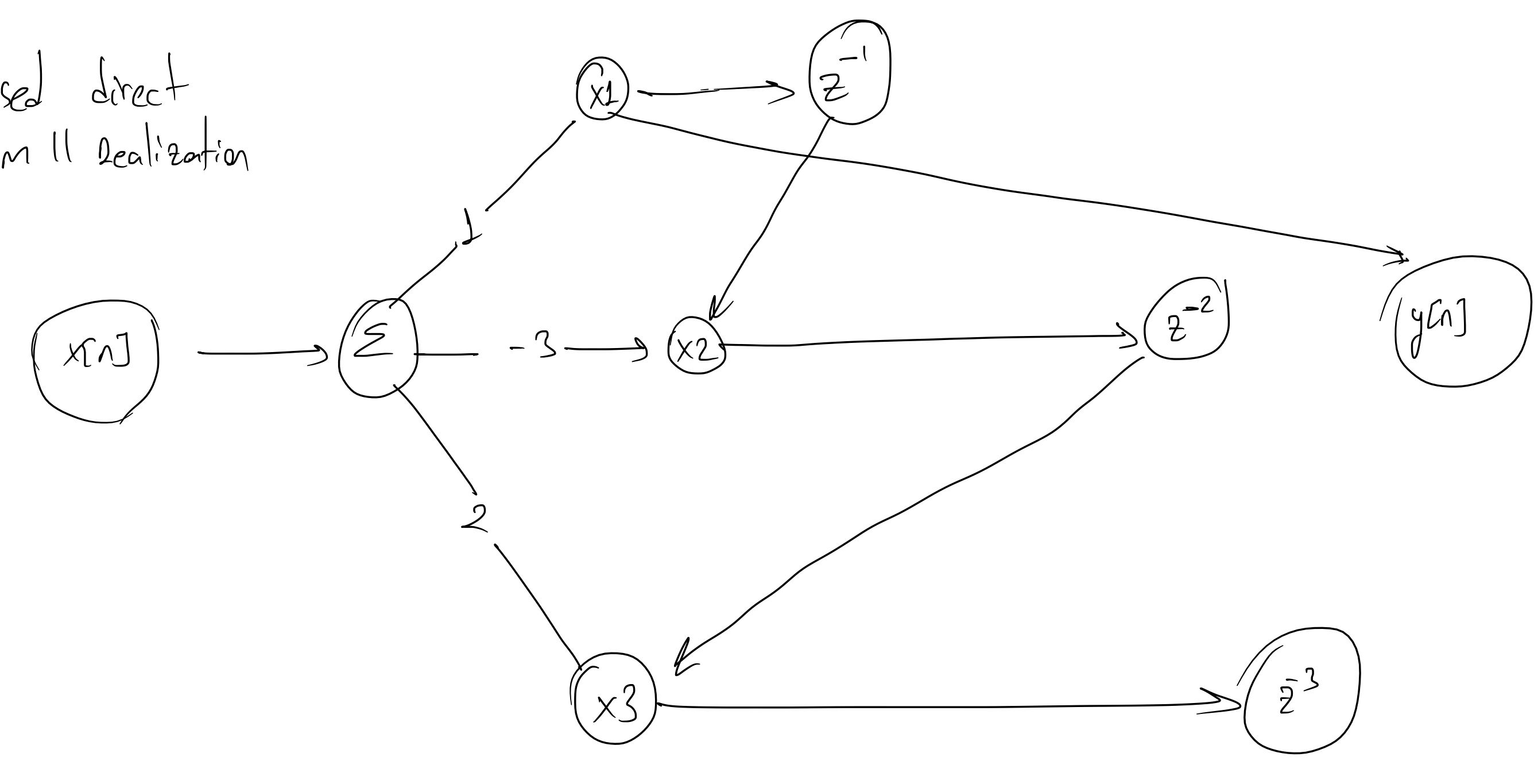
3 gecikme bloğu: z^{-1}, z^{-2}, z^{-3}

feed forward katsayı : 1, −3, 2

feedback kat sayı : 3, −4, −2

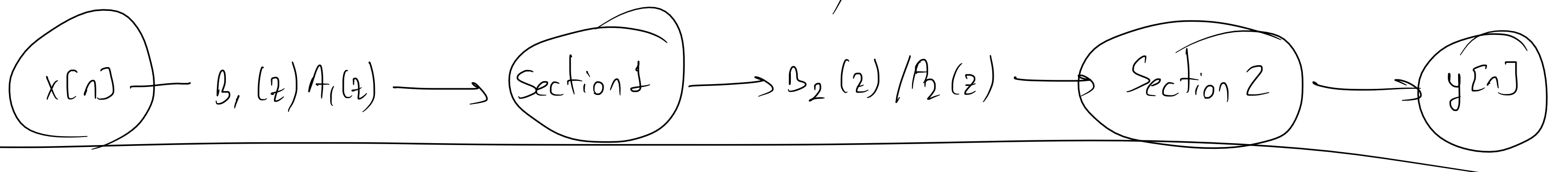
2)

Transposed direct form II Realization

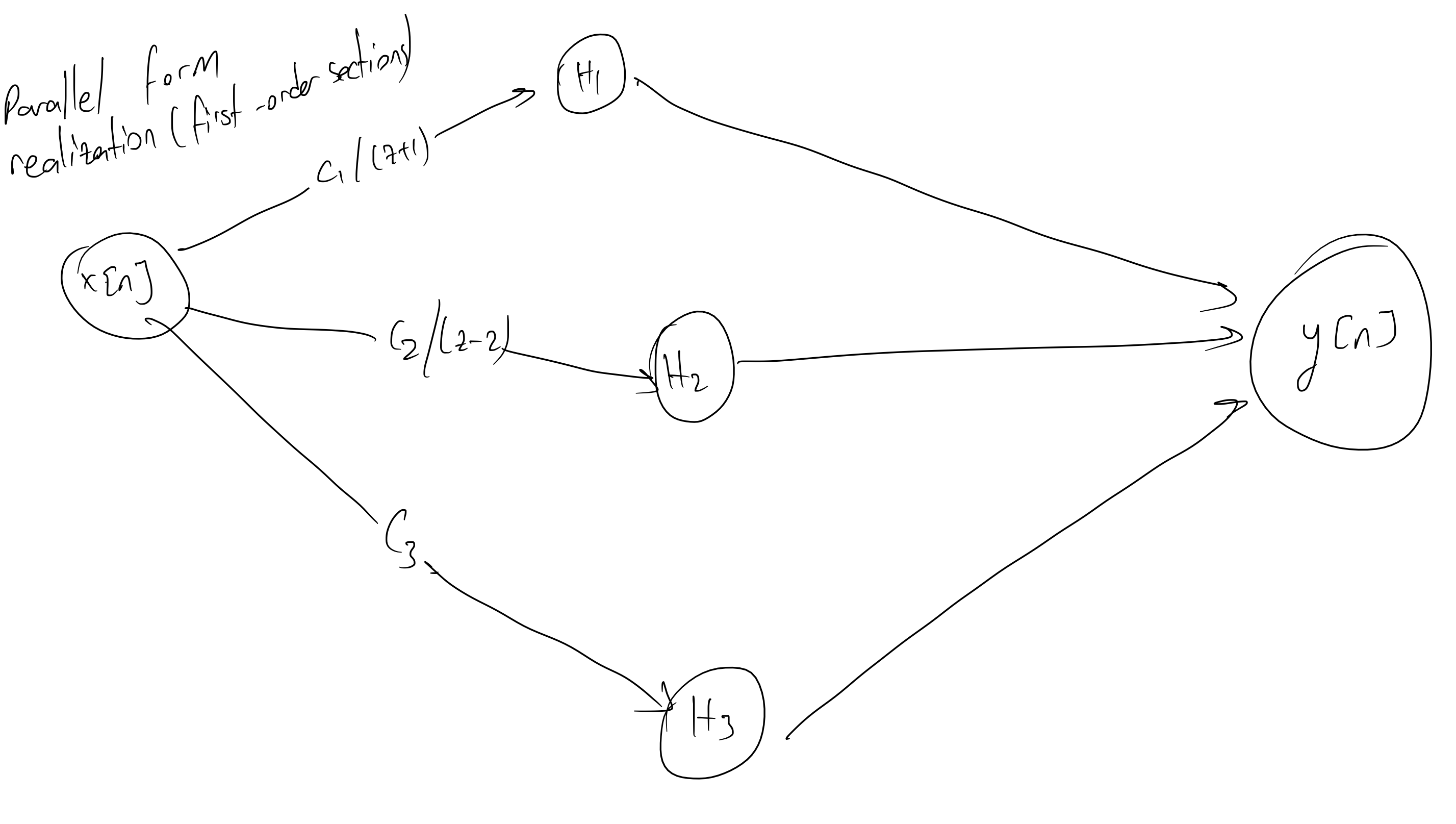


3)

cascade form realization (Two second-order sections)



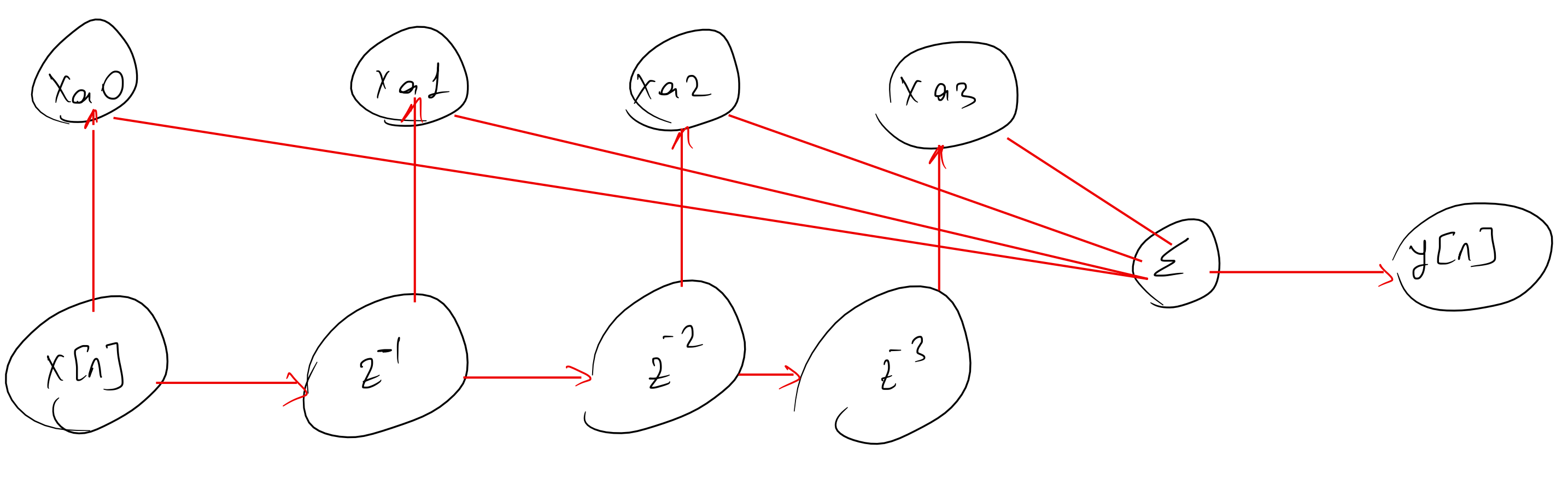
b) Parallel form realization (first-order sections)



$$H(z) = \frac{z(z-1)}{(z+1)(z-2)} \Rightarrow \underbrace{\frac{c_1}{z+1} + \frac{c_2}{z-2}}_{\text{birer kutuplu (first order) filtre}} + \underbrace{c_3}_{\text{sabit kazanç}}$$

---

c)



This direct from implementation of an FIR filter. The input signal passes through a series of delay elements, and each delayed sample is multiplied by a corresponding coefficient. The outputs of these multipliers are summed to produce final output, y[n]. Since there is no feedback, the structure is stable and purely feedforward.

Hocam ilk ödevin teslim tarihini başka bir tarihle karıştırdığım için yaptığım halde yükleyememiştim. Aşağıda ödevimin o tarihte tamamlandığı kadarıyla gönderiyorum. Kabul ederseniz çok sevinirim, notumu yükseltmeye çalışıyorum. Hiçbir ekleme yapılmamıştır. Teşekkürler.

Mehmet Ali Kamalak

Homework 1

Fourier dönüşümü

$x[n] = X(e^{j\omega})$

$x[n-1] = e^{-j\omega} X(e^{j\omega})$

Question 1

An LTI system has the frequency response

$$H(e^{j\omega}) = \frac{1 - 1.25e^{-j\omega}}{1 - 0.8e^{-j\omega}} = 1 - \frac{0.45e^{-j\omega}}{1 - 0.8e^{-j\omega}} \qquad (1)$$

(a) Specify the difference equation that is satisfied by the input $x[n]$ and the output $y[n]$.

(b) Use one of the above forms of the frequency response to determine the impulse response $h[n]$.

(c) Show that $|H(e^{j\omega})|^2 = G^2$, where G is a constant. Determine the constant G. (This is an example of an allpass filter.)

(d) If the input to the above system is $x[n] = cos(0.2\pi n)$, the output should be of the form $y[n] = Acos(0.2\pi n + \theta)$. What are A and θ?

$$\boxed{e^{-j\omega} = z^{-1}}$$

$$H(e^{j\omega}) = \frac{1 - 1{,}25e^{-j\omega}}{1 - 0{,}8e^{-j\omega}} = 1 - \frac{0{,}45e^{-j\omega}}{1 - 0{,}8e^{-j\omega}}$$

a) Difference equation for input x[n] and output y[n]

$1{,}25z^{-1} = x[n-1]$

$1 = x[n]$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 - 1{,}25z^{-1}}{1 - 0{,}8z^{-1}} \Rightarrow Y(z)(1 - 0{,}8z^{-1}) = X(z)(1 - 1{,}25z^{-1})$$

$$\boxed{y[n] - 0{,}8\,y[n-1] = x[n] - 1{,}25\,x[n-1]}$$

b) Impulse response one of the form

I choose

$$H(e^{j\omega}) = 1 - \frac{0{,}45e^{-j\omega}}{1 - 0{,}8e^{-j\omega}}$$

$$\frac{1}{1 - az^{-1}} = \frac{1}{1 - ae^{-j\omega}} \Leftrightarrow a^n u[n]$$

$e^{-j\omega} \Leftrightarrow$ gecikme

$$\frac{z^{-1}}{(1 - az^{-1})^2} \overset{z^{-1}}{\Rightarrow} n \cdot a^{n-1} u[n-1]$$

$$\underbrace{1 - 0{,}45 \cdot}\ \frac{e^{-j\omega}}{\underbrace{1 - 0{,}8e^{-j\omega}}} \overset{z^{-1}}{\Rightarrow}$$

z transformu 1'di

$$h[n] = \delta[n] - 0{,}45\left(n \cdot (0{,}8^{n-1}) \cdot u[n-1]\right)$$

c) $||H(e^{j\omega})|^2| = G^2$ G is constant

$e^{-j\omega} = (\cos\omega - j\sin\omega)$

$$H(e^{j\omega}) = \frac{1 - 1{,}25e^{-j\omega}}{1 - 0{,}8e^{-j\omega}}$$

$|A|^2 = A \cdot A^*$ $\underline{\underline{1 + e^{-j\omega}}}$ → $\underline{\underline{1 + e^{j\omega}}}$

$$= \left(\frac{1 - 1{,}25e^{-j\omega}}{1 - 0{,}8e^{-j\omega}}\right) \cdot \left(\frac{1 - 1{,}25e^{j\omega}}{1 - 0{,}8e^{j\omega}}\right)$$

$$= \frac{1 - 1{,}25e^{j\omega} - 1{,}25e^{-j\omega} + (1{,}25)^2}{1 - 0{,}8e^{-j\omega} - 0{,}8e^{j\omega} + (0{,}8)^2} = \frac{2{,}56 - 1{,}25(e^{j\omega} + e^{-j\omega})}{1{,}64 - 0{,}8(e^{j\omega} + e^{-j\omega})} = ||H(e^{j\omega})|^2|$$

$$\frac{1{,}25\left(2 - \cancel{(e^{j\omega} + e^{-j\omega})}\right)}{0{,}8\left(2 - \cancel{(e^{j\omega} + e^{-j\omega})}\right)} = 1{,}5625 = G^2$$

$$\boxed{G = 1{,}25}$$

d) if the input $x[n] = \overset{\cos(\omega n)}{\cos(0{,}2\pi n)}$ output should be of form $y[n] = A\cos(0{,}2\pi n + \theta)$. What are A and θ?

$$H(e^{j\omega}) = \frac{1 - 1{,}25e^{-j\omega}}{1 - 0{,}8e^{-j\omega}} \Rightarrow \omega = 2\pi$$

$$e^{-j\omega} = e^{-j(0{,}2\pi)} = \cos(\underset{36^\circ}{\underbrace{0{,}2\pi}}) - j\sin(\underset{36^\circ}{\underbrace{0{,}2\pi}})$$

$$\frac{1 - 1{,}25\left(\overbrace{\cos(0{,}2\pi)}^{0{,}8090} - j\overbrace{\sin(0{,}2\pi)}^{0{,}5878}\right)}{1 - 0{,}8\left(\cos(0{,}2\pi) - j\sin(0{,}2\pi)\right)} = \frac{1 - 1{,}011 + j0{,}735}{1 - 0{,}6472 + j0{,}4702} = \frac{-0{,}011 + j0{,}735}{0{,}3528 + j0{,}4702} = A\cos[0{,}2\pi n + \theta]$$

$$A = |H(e^{j\omega_0})| = \frac{\sqrt{(0{,}011)^2 + (0{,}735)^2}}{\sqrt{(0{,}3528)^2 + (0{,}4702)^2}} = \frac{0{,}735}{0{,}588} = \underline{1{,}25} = A$$

$\theta = \arg(H(e^{j\omega_0}))$

$$\theta = \arg\left(\frac{1 - 1{,}25e^{-j0{,}2\pi}}{1 - 0{,}8e^{-j0{,}2\pi}}\right)$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{1 - 1{,}25e^{-j\omega}}{1 - 0{,}8e^{-j\omega}}$$

a) $X(e^{j\omega}) \cdot (1 - 1{,}25 \cdot e^{-j\omega}) = Y(e^{j\omega}) \cdot (1 - 0{,}8e^{-j\omega})$

$X(e^{j\omega}) - X(e^{j\omega}) \cdot 1{,}25 \cdot e^{-j\omega} =$

$x[n] - x[n-1] \cdot 1{,}25 = y[n] - 0{,}8 \cdot y[n-1]$

$$\boxed{y[n] = 0{,}8\,y[n-1] + x[n] - 1{,}25\,x[n-1]}$$

b) $H(e^{j\omega}) = 1 - \dfrac{0{,}45e^{-j\omega}}{1 - 0{,}8e^{-j\omega}}$ $\Rightarrow u[n-1] \cdot a^{n-1} \rightarrow \dfrac{a \cdot e^{-j\omega}}{1 - a \cdot e^{-j\omega}}$ ✓

$$h[n] = \delta[n] - 0{,}45(0{,}8)^{n-1} \cdot u[n-1]$$

c) $|H(e^{j\omega})|^2 = G^2$ $||H(e^{j\omega})|^2| = H(e^{j\omega}) \cdot H^*(e^{j\omega})$ ↳ complex conjugate (eşlenik) ✓

$$H(e^{j\omega}) = \frac{1 - 1{,}25e^{-j\omega}}{1 - 0{,}8e^{-j\omega}}$$

$$H^*(e^{j\omega}) = \frac{1 - 1{,}25e^{j\omega}}{1 - 0{,}8e^{j\omega}} \qquad \boxed{(e^{-j\omega})^* = e^{j\omega}}$$ ✓

$$|H(e^{j\omega})|^2 = \frac{1 - 1{,}25e^{-j\omega}}{1 - 0{,}8e^{-j\omega}} \cdot \frac{1 - 1{,}25e^{j\omega}}{1 - 0{,}8e^{j\omega}} = \frac{1 - 1{,}25(e^{-j\omega} + e^{j\omega}) + (1{,}25)^2}{(1 - 0{,}8e^{-j\omega}) \cdot (1 - 0{,}8e^{j\omega})}$$

$$= \frac{1 - 2{,}5(\cos\omega) + 1{,}5625}{1 - 1{,}6\cos\omega + 0{,}64}$$

$$G^2 = \frac{2{,}5625(1 - 0{,}9756\cos(\omega))}{1{,}64(1 - 0{,}9756\cos(\omega))} = \frac{2{,}5625}{1{,}64} = (1{,}25)^2$$

G = 1,25   All pass filter olması için genlik 1 olmalıydı ✓

Not all pass filter

d) $x[n] = \cos(0{,}2\pi n)$

$y[n] = A\cos(0{,}2\pi n + \theta)$

$y[n] =$

↳ input (x(n)) $\cos(\omega n) = \mathrm{Re}(e^{j\omega n})$

output (y(n))

↳ $|H(e^{j\omega})| \cdot \cos(\omega \cdot n + \angle H(e^{j\omega}))$ ✓

$\omega = 0{,}2\pi$ $\theta = \angle H(e^{j0{,}2\pi})$

$|H(e^{j0{,}2\pi})| = A$ ✓

$$H(e^{j\omega}) = \frac{1 - 1{,}25e^{-j\omega}}{1 - 0{,}8e^{-j\omega}} \Rightarrow e^{-j\omega} = \cos(\omega) - j\sin\omega$$

$$H(e^{j0{,}2\pi}) = \frac{1 - 1{,}25e^{-j0{,}2\pi}}{1 - 0{,}8e^{-j0{,}2\pi}} \qquad e^{-j0{,}2\pi} = \cos(0{,}2\pi) - j\sin(0{,}2\pi) = 0{,}99 - 0{,}01j$$

$$= \frac{1 - 1{,}25(0{,}99 - 0{,}01j)}{1 - 0{,}8(0{,}99 - 0{,}01j)} = \frac{-0{,}24 + 0{,}01j}{0{,}208 + 0{,}008j} =$$

$$|H(e^{j0{,}2\pi})| = \frac{\sqrt{(0{,}24)^2 + (0{,}01)^2}}{\sqrt{(0{,}208)^2 + (0{,}008)^2}} = 1{,}15 = A$$

$$\theta \Rightarrow H(e^{j0{,}2\pi}) = \frac{-0{,}24 + 0{,}01j}{0{,}208 + 0{,}008j} \Rightarrow$$

$\theta_{pay} = \tan^{-1}\left(\dfrac{0{,}01}{-0{,}24}\right) \approx \dfrac{\pi}{2}$

$\theta_{payda} = \tan^{-1}\left(\dfrac{0{,}008}{0{,}208}\right) = 0{,}93\,\mathrm{rad} = 0{,}296\pi$

açı farkını bul.

$\theta = \angle H(e^{j\omega}) = \dfrac{\pi}{2} - 0{,}296\pi = \underline{0{,}204\pi}$ rad ✓

$$A\cos(0{,}2\pi n + \theta) = \underline{1{,}15\cos(0{,}2\pi n + 0{,}204\pi)}$$

---

66. The autocorrelation sequence of a signal $x[n]$ is defined as

$$R_x[n] = \sum_{k=-\infty}^{\infty} x^*[k]x[n+k].$$

(a) Show that for an appropriate choice of the signal $g[n]$, $R_x[n] = x[n] * g[n]$, and identify the proper choice for $g[n]$.
(b) Show that the Fourier transform of $R_x[n]$ is equal to $|X(e^{j\omega})|^2$.

a) $r = -k$

$$\sum_{r=-\infty}^{\infty} x^*[-r] \cdot x[n-r] \rightarrow \underline{\underline{x^*[-n] * x[n]}}$$

Therefore

$$\underline{g[n] = x^*[-n]}$$

b) Fourier transform of $x^*[-n]$ is $X^*(e^{j\omega})$, therefore

$$R_x(e^{j\omega}) = X^*(e^{j\omega}) \cdot X(e^{j\omega}) = |X(e^{j\omega})^2|$$